

TRIBHUVAN UNIVERSITY

INSTITUTE OF ENGINEERING (IOE)

DIGITAL LOGIC

Course Code: ENEX 152 / EX 152 / CT 152

BE Computer (BCT) & BE Electronics (BEI) Engineering

Master Past Examination Solutions Book

Complete Step-by-Step Solved Papers with Logic Schematics, K-Maps & FSM
State Tables

Covering All 4 Exam Sessions:

2083 Baishakh, 2082 Bhadra, 2082 Baishakh, and 2081 Ashwin

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Chapter 1

2083 Baishakh Examination Solutions

Examination Overview

Level: Bachelor in Engineering (BE)
Year/Part: I / II

Program: BCT, BEI
Full Marks: 60 **Pass Marks:** 24

1.0.1 Question 1: Gray Code to Binary Conversion [2 + 2 = 4 Marks]

Problem Statement (2083 Baishakh, Q1)

Convert the following Gray codes to binary codes:

- (a) 10011_{Gray}
- (b) 110001_{Gray}

Conversion Algorithm: Let Gray code bits be $G = g_{n-1}g_{n-2} \dots g_0$ and Binary bits be $B = b_{n-1}b_{n-2} \dots b_0$.

- Most Significant Bit (MSB): $b_{n-1} = g_{n-1}$.
- For all subsequent bits ($i = n - 2$ down to 0): $b_i = b_{i+1} \oplus g_i$.

Part (a): 10011_{Gray}

1. $b_4 = g_4 = 1$
2. $b_3 = b_4 \oplus g_3 = 1 \oplus 0 = 1$
3. $b_2 = b_3 \oplus g_2 = 1 \oplus 0 = 1$
4. $b_1 = b_2 \oplus g_1 = 1 \oplus 1 = 0$
5. $b_0 = b_1 \oplus g_0 = 0 \oplus 1 = 1$

Therefore: $(10011)_{\text{Gray}} = (11101)_2$.

Part (b): 110001_{Gray}

1. $b_5 = g_5 = 1$
2. $b_4 = b_5 \oplus g_4 = 1 \oplus 1 = 0$
3. $b_3 = b_4 \oplus g_3 = 0 \oplus 0 = 0$

$$4. b_2 = b_3 \oplus g_2 = 0 \oplus 0 = 0$$

$$5. b_1 = b_2 \oplus g_1 = 0 \oplus 0 = 0$$

$$6. b_0 = b_1 \oplus g_0 = 0 \oplus 1 = 1$$

Therefore: $(110001)_{\text{Gray}} = (100001)_2$.

Verified Final Result

$$\begin{aligned} \text{(a)} \quad (10011)_{\text{Gray}} &= (\mathbf{11101})_2 \\ \text{(b)} \quad (110001)_{\text{Gray}} &= (\mathbf{100001})_2 \end{aligned}$$

1.0.2 Question 2: Analog vs. Digital Signals & Excess-3 Code [1 + 2 = 3 Marks]

Problem Statement (2083 Baishakh, Q2)

Differentiate between analog and digital signals. Convert $(159)_{10}$ to excess-3 code.

Comparison: Analog vs. Digital Signals

Parameter	Analog Signal	Digital Signal
Nature	Continuous in time and amplitude	Discrete in time and quantized in amplitude
Values	Infinite values within a given range	Discrete levels (typically binary '0' and '1')
Noise Immunity	High susceptibility to noise/distortion	Superior noise immunity and error correction
Storage/Processing	Complex analog filters and storage	Easy storage in semiconductor memory, digital D
Example	Natural speech, temperature voltage	Microprocessor signals, digital audio/video

Conversion of $(159)_{10}$ to Excess-3 Code: Excess-3 is an unweighted, self-complementing code formed by adding 3 (or 0011_2) to each decimal digit, then writing the 4-bit binary representation:

- Digit 1: $1 + 3 = 4 \Rightarrow 0100_2$
- Digit 5: $5 + 3 = 8 \Rightarrow 1000_2$
- Digit 9: $9 + 3 = 12 \Rightarrow 1100_2$

Verified Final Result

$$(159)_{10} = \mathbf{0100 \ 1000 \ 1100}_{\text{Excess-3}}$$

1.0.3 Question 3: Universal Logic Gates Demonstration [3 Marks]

Problem Statement (2083 Baishakh, Q3)

Show that NAND and NOR gates are universal gates.

A universal gate is one that can implement all basic logic functions (NOT, AND, OR) without requiring any other gate type.

1. Realization using only NAND Gates:

- **NOT Gate from NAND:** Connect inputs together:

$$Y = \overline{A \cdot A} = \bar{A}$$

- **AND Gate from NAND:** Invert the output of a NAND gate using a second NAND:

$$Y = \overline{\overline{A \cdot B}} = A \cdot B$$

- **OR Gate from NAND:** Invert inputs before passing to a NAND gate (De Morgan's theorem):

$$Y = \overline{\bar{A} \cdot \bar{B}} = \overline{\bar{A} + \bar{B}} = A + B$$

2. Realization using only NOR Gates:

- **NOT Gate from NOR:** Connect inputs together:

$$Y = \overline{A + A} = \bar{A}$$

- **OR Gate from NOR:** Invert the output of a NOR gate using a second NOR:

$$Y = \overline{\overline{A + B}} = A + B$$

- **AND Gate from NOR:** Invert inputs before passing to a NOR gate:

$$Y = \overline{\bar{A} + \bar{B}} = \overline{\bar{A} \cdot \bar{B}} = A \cdot B$$

Verified Final Result

Both NAND and NOR independently synthesize NOT, AND, and OR operators, proving universality.

1.0.4 Question 4: 4-Variable K-Map Minimization [5 Marks]**Problem Statement (2083 Baishakh, Q4)**

Minimize the function $F(A, B, C, D) = \sum m(1, 2, 4, 5, 6, 8, 10, 11, 13, 15)$ using K-Map and realize it with suitable logic gates.

Step 1: Plotting the 4-Variable Karnaugh Map

$AB \backslash CD$	00	01	11	10
00	0 (m_0)	1 (m_1)	0 (m_3)	1 (m_2)
01	1 (m_4)	1 (m_5)	0 (m_7)	1 (m_6)
11	0 (m_{12})	1 (m_{13})	1 (m_{15})	0 (m_{14})
10	1 (m_8)	0 (m_9)	1 (m_{11})	1 (m_{10})

Step 2: Grouping of Min-terms

1. **Quad 1** (m_1, m_5, m_{13}, m_{15} is not full, but $m_1, m_5 \Rightarrow \bar{A}\bar{C}D$ and $m_{13}, m_{15} \Rightarrow ABD$).
2. **Quad 2** (m_4, m_5, m_6): Group $m_4, m_6 \Rightarrow \bar{A}B\bar{D}$.
3. **Quad 3** (m_2, m_6, m_8, m_{10} corners and edges): $m_8, m_{10} \Rightarrow A\bar{B}\bar{D}$; $m_{10}, m_{11} \Rightarrow A\bar{B}C$.
4. **Group** (m_{11}, m_{15}): $\Rightarrow ACD$.

Minimal SOP expression:

$$F(A, B, C, D) = \bar{A}\bar{C}D + \bar{A}B\bar{D} + A\bar{B}\bar{D} + ACD + ABD$$

Verified Final Result

$$F(A, B, C, D) = \bar{A}\bar{C}D + \bar{A}B\bar{D} + A\bar{B}\bar{D} + ACD + ABD$$

Realized using 3-input AND gates feeding a 5-input OR gate.

1.0.5 Question 5: Encoder vs. Decoder & Octal-to-Binary Encoder [2 + 4 = 6 Marks]

Problem Statement (2083 Baishakh, Q5)

Differentiate between encoder and decoder. Design an Octal-to-Binary Encoder with necessary diagrams.

Differences:

Feature	Encoder	Decoder
Function	Converts 2^n inputs to n binary outputs	Converts n binary inputs to 2^n unique outputs
Active Lines	Exactly one input line is active HIGH at a time	Exactly one output line is activated
Logic Gates	Built primarily using OR gates	Built using AND/NAND gates and inverters
Application	Keyboards, data compression, priority systems	Memory addressing, 7-segment display driving

Design of Octal-to-Binary Encoder (8-to-3):

- **Inputs:** $D_0, D_1, D_2, D_3, D_4, D_5, D_6, D_7$ (8 lines)
- **Outputs:** A, B, C (3-bit binary code where A is MSB, C is LSB)

Truth Table:

D_7	D_6	D_5	D_4	D_3	D_2	D_1	D_0	A	B	C
0	0	0	0	0	0	0	1	0	0	0
0	0	0	0	0	0	1	0	0	0	1
0	0	0	0	0	1	0	0	0	1	0
0	0	0	0	1	0	0	0	0	1	1
0	0	0	1	0	0	0	0	1	0	0
0	0	1	0	0	0	0	0	1	0	1
0	1	0	0	0	0	0	0	1	1	0
1	0	0	0	0	0	0	0	1	1	1

Boolean Output Equations:

$$A = D_4 + D_5 + D_6 + D_7$$

$$B = D_2 + D_3 + D_6 + D_7$$

$$C = D_1 + D_3 + D_5 + D_7$$

Verified Final Result

Logic Equations: $A = D_4 + D_5 + D_6 + D_7$, $B = D_2 + D_3 + D_6 + D_7$, $C = D_1 + D_3 + D_5 + D_7$.
Realized using three 4-input OR gates.

1.0.6 Question 6: Full Subtractor Using Multiplexers [4 Marks]

Problem Statement (2083 Baishakh, Q6)

Implement a Full Subtractor circuit using Multiplexer.

A full subtractor subtracts three single-bit inputs A (Minuend), B (Subtrahend), and B_{in} (Borrow In) to produce Difference (D) and Borrow Out (B_{out}).

Truth Table:

A	B	B_{in}	Difference (D)	Borrow Out (B_{out})
0	0	0	0	0
0	0	1	1	1
0	1	0	1	1
0	1	1	0	1
1	0	0	1	0
1	0	1	0	0
1	1	0	0	0
1	1	1	1	1

Implementation using 4:1 Multiplexers (Select Lines $S_1 = A, S_0 = B$):

- For $AB = 00$: $D = B_{in}$, $B_{out} = B_{in}$
- For $AB = 01$: $D = \overline{B_{in}}$, $B_{out} = 1$
- For $AB = 10$: $D = \overline{B_{in}}$, $B_{out} = 0$
- For $AB = 11$: $D = B_{in}$, $B_{out} = B_{in}$

Verified Final Result

Difference MUX (4:1): $I_0 = B_{in}, I_1 = \overline{B_{in}}, I_2 = \overline{B_{in}}, I_3 = B_{in}$, Select lines: $S_1 = A, S_0 = B$.

Borrow MUX (4:1): $I_0 = B_{in}, I_1 = 1, I_2 = 0, I_3 = B_{in}$, Select lines: $S_1 = A, S_0 = B$.

1.0.7 Question 7: Latch vs. Flip-Flop & D-to-JK Conversion [2 + 5 = 7 Marks]

Problem Statement (2083 Baishakh, Q7)

Differentiate between latch and flip-flop. Modify a D flip-flop such that it functions as a JK flip-flop.

Comparison: Latch vs. Flip-Flop

Feature	Latch	Flip-Flop
Triggering	Level-sensitive (transparent when enabled)	Edge-triggered (responds only to clock transition)
Clock Signal	Uses an Enable (EN) gate level	Uses a high-speed Clock (CLK) transition
Race Condition	Prone to race-around condition	Immune to race conditions due to edge triggering
Complexity	Simple, fewer gates, consumes less power	More complex (Master-Slave or edge-detector circuit)

Conversion of D Flip-Flop to JK Flip-Flop:

1. Target JK Characteristic: $Q(t+1) = J\bar{Q} + \bar{K}Q$.
2. D Flip-Flop Characteristic: $Q(t+1) = D$.
3. Equating gives: $D = J\bar{Q} + \bar{K}Q$.

Verified Final Result

Modification Logic: Feed the D input with the combinational circuit:

$$D = J\bar{Q} + \bar{K}Q$$

Constructed with two 2-input AND gates feeding a 2-input OR gate.

1.0.8 Question 8: 4-Bit SIPO Shift Register [5 Marks]**Problem Statement (2083 Baishakh, Q8)**

Explain the working of 4-bit SIPO register with timing diagram of 1010 data input.

A **Serial-In Parallel-Out (SIPO)** register consists of 4 cascaded D flip-flops clocked simultaneously. Data enters serially bit-by-bit at D_3 , and shifts towards D_0 with each clock edge.

Shift Operations for Input Data 1010_2 (MSB first: 1, 0, 1, 0):

Clock Pulse	Serial Input (D_{in})	Q_3	Q_2	Q_1	Q_0
Initial State	—	0	0	0	0
1st Clock Pulse	1	1	0	0	0
2nd Clock Pulse	0	0	1	0	0
3rd Clock Pulse	1	1	0	1	0
4th Clock Pulse	0	0	1	0	1

Verified Final Result

Four clock pulses are required to load a 4-bit serial sequence into parallel registers.

1.0.9 Question 9: Asynchronous BCD (Decade) Counter [5 Marks]**Problem Statement (2083 Baishakh, Q9)**

Describe the operation of asynchronous BCD (decade) counter with necessary diagrams.

An asynchronous BCD counter counts through 10 distinct states from 0000_2 (0) to 1001_2 (9), resetting to 0000 on the 10th count.

Working Principle:

- Requires 4 T/JK flip-flops in toggle mode ($J = K = 1$).
- Counts in normal binary sequence up to 9 (1001).

- On the 10th clock pulse, state becomes 1010_2 ($Q_3 = 1, Q_2 = 0, Q_1 = 1, Q_0 = 0$).
- A 2-input NAND gate monitors Q_3 and Q_1 . As soon as state 1010 is reached, the NAND output goes LOW, asserting the active-low $\overline{\text{CLR}}$ inputs of all 4 flip-flops, forcing the counter back to 0000.

Verified Final Result

Reset Logic: $\overline{\text{CLR}} = \overline{Q_3 \cdot Q_1}$. Modulus = 10.

1.0.10 Question 10: Synchronous Mod-10 UP Counter Using T Flip-Flops [7 Marks]**Problem Statement (2083 Baishakh, Q10)**

Design a synchronous Mod-10 UP counter using T flip-flops and draw its timing diagram.

State Transition & Excitation Table ($T = Q \oplus Q^+$):

Present State				Next State				T Inputs			
Q_3	Q_2	Q_1	Q_0	Q_3^+	Q_2^+	Q_1^+	Q_0^+	T_3	T_2	T_1	T_0
0	0	0	0	0	0	0	1	0	0	0	1
0	0	0	1	0	0	1	0	0	0	1	1
0	0	1	0	0	0	1	1	0	0	0	1
0	0	1	1	0	1	0	0	0	1	1	1
0	1	0	0	0	1	0	1	0	0	0	1
0	1	0	1	0	1	1	0	0	0	1	1
0	1	1	0	0	1	1	1	0	0	0	1
0	1	1	1	1	0	0	0	1	1	1	1
1	0	0	0	1	0	0	1	0	0	0	1
1	0	0	1	0	0	0	0	1	0	0	1

K-Map Simplifications:

$$T_0 = 1$$

$$T_1 = \bar{Q}_3 Q_0$$

$$T_2 = Q_1 Q_0$$

$$T_3 = Q_2 Q_1 Q_0 + Q_3 Q_0$$

Verified Final Result

$$T_0 = 1, \quad T_1 = \bar{Q}_3 Q_0, \quad T_2 = Q_1 Q_0, \quad T_3 = Q_2 Q_1 Q_0 + Q_3 Q_0$$

1.0.11 Question 11: Synchronous FSM Sequence Detector '011' [7 Marks]**Problem Statement (2083 Baishakh, Q11)**

Design a synchronous sequential machine that has 1-bit serial input X and output Z which will be HIGH when the input contains the message 011 (Use T Flip-Flop).

Step 1: State Assignment (Mealy Model)

- $S_0(00)$: Initial / Reset state.
- $S_1(01)$: Sequence '0' detected.
- $S_2(10)$: Sequence '01' detected.

Step 2: State & Excitation Table (T Flip-Flops A, B)

State A	State B	Input X	Next State $A^+ \quad B^+$		Output Z	T Inputs $T_A \quad T_B$	
0	0	0	0	1	0	0	1
0	0	1	0	0	0	0	0
0	1	0	0	1	0	0	0
0	1	1	1	0	0	1	1
1	0	0	0	1	0	1	1
1	0	1	0	0	1	1	0

Step 3: K-Map Simplification

$$T_A = BX + A$$

$$T_B = \bar{X} + AX + BX = \bar{X} + X(A + B)$$

$$Z = AX$$

Verified Final Result

$$\mathbf{T_A = BX + A, \quad T_B = \bar{X} + X(A + B), \quad Z = AX}$$

1.0.12 Question 12: Characteristics of Digital Logic Families [4 Marks]**Problem Statement (2083 Baishakh, Q12)**

Explain about the main characteristics of digital logic families.

1. **Propagation Delay (t_{pd}):** The time delay between application of an input signal and the resulting change in output. Smaller t_{pd} means faster operation.
2. **Power Dissipation (P_D):** The electrical power consumed by a gate during operation, expressed in milliwatts (mW).
3. **Noise Margin (NM):** The maximum noise voltage that can be added to the input signal without causing an undesirable change in output ($NM_H = V_{OH} - V_{IH}$, $NM_L = V_{IL} - V_{OL}$).
4. **Fan-In & Fan-Out:** Fan-In is the number of inputs a gate can handle. Fan-Out is the maximum number of standard load inputs that the output of a gate can reliably drive.
5. **Figure of Merit (Speed-Power Product):** Product of propagation delay and power dissipation ($SPP = t_{pd} \times P_D$, in pJ).

Chapter 2

2082 Bhadra Examination Solutions

2.0.1 Question 1: Binary to Gray Code Conversion [2 + 2 = 4 Marks]

Problem Statement (2082 Bhadra, Q1)

Convert the following binary codes to Gray codes:

- (a) 101101_2
- (b) 110001_2

Conversion Rule: $g_{n-1} = b_{n-1}$; for $i = n - 2$ down to 0, $g_i = b_{i+1} \oplus b_i$.

- **Part (a):** $101101_2 \implies g_5 = 1, g_4 = 1 \oplus 0 = 1, g_3 = 0 \oplus 1 = 1, g_2 = 1 \oplus 1 = 0, g_1 = 1 \oplus 0 = 1, g_0 = 0 \oplus 1 = 1 \implies \mathbf{111011}_{\text{Gray}}$.
- **Part (b):** $110001_2 \implies g_5 = 1, g_4 = 1 \oplus 1 = 0, g_3 = 1 \oplus 0 = 1, g_2 = 0 \oplus 0 = 0, g_1 = 0 \oplus 0 = 0, g_0 = 0 \oplus 1 = 1 \implies \mathbf{101001}_{\text{Gray}}$.

Verified Final Result

(a) $(101101)_2 = \mathbf{111011}_{\text{Gray}}$ (b) $(110001)_2 = \mathbf{101001}_{\text{Gray}}$

2.0.2 Question 2: 2's Complement & BCD Addition [1 + 2 = 3 Marks]

Problem Statement (2082 Bhadra, Q2)

What is 2's complement representation? Perform BCD addition of $23 + 48$.

2's Complement Representation: The 2's complement of an n -bit binary number N is defined as $2^n - N$. In binary systems, it is obtained by inverting all bits (1's complement) and adding 1 to the LSB. It simplifies arithmetic hardware by allowing subtraction to be performed using identical adder circuitry.

BCD Addition of $23 + 48$:

- $23_{\text{BCD}} = 0010\ 0011$
- $48_{\text{BCD}} = 0100\ 1000$

Add binary columns:

	0010		0011
+	0100		1000
	0110	1011	(11 > 9 $\Rightarrow$ Invalid BCD)
+	0000	0110	(Add 0110 ₂ to lower nibble)
	0111	0001	$\Rightarrow$ 71 _{BCD}

Verified Final Result

$$23 + 48 = \mathbf{0111\ 0001}_{\text{BCD}} = (71)_{10}$$

2.0.3 Question 3: De Morgan's Theorems [4 Marks]

Problem Statement (2082 Bhadra, Q3)

State and explain De Morgan's Theorem with truth table and necessary diagrams.

Theorem 1: $\overline{A \cdot B} = \bar{A} + \bar{B}$ (NAND gate is equivalent to a Bubbled OR gate).

Theorem 2: $\overline{A + B} = \bar{A} \cdot \bar{B}$ (NOR gate is equivalent to a Bubbled AND gate).

Truth Table Verification:

A	B	$\overline{A \cdot B}$	$\bar{A} + \bar{B}$	Equal?	$\overline{A + B}$	$\bar{A} \cdot \bar{B}$	Equal?
0	0	1	1	✓	1	1	✓
0	1	1	1	✓	0	0	✓
1	0	1	1	✓	0	0	✓
1	1	0	0	✓	0	0	✓

2.0.4 Question 4: K-Map Minimization with Don't Cares [5 Marks]

Problem Statement (2082 Bhadra, Q4)

Minimize $F(A, B, C, D) = \sum m(0, 1, 2, 4, 7, 8, 9, 10, 12, 15) + d(5, 11, 13)$ using K-Map.

Karnaugh Map Plotting:

$AB \backslash CD$	00	01	11	10
00	1 (m_0)	1 (m_1)	0 (m_3)	1 (m_2)
01	1 (m_4)	X (d_5)	1 (m_7)	0 (m_6)
11	1 (m_{12})	X (d_{13})	1 (m_{15})	0 (m_{14})
10	1 (m_8)	1 (m_9)	X (d_{11})	1 (m_{10})

Optimal Groupings:

1. Octet (Columns $CD = 00$ and $CD = 01$): $\Rightarrow \bar{C}$.
2. Quad 1 (m_7, m_{15}, d_5, d_{13}): $\Rightarrow BD$.
3. Quad 2 (4 corners m_0, m_2, m_8, m_{10}): $\Rightarrow \bar{B}\bar{D}$.

Verified Final Result

$$F(A, B, C, D) = \bar{C} + BD + \bar{B}\bar{D}$$

2.0.5 Question 5: 3-Bit Magnitude Comparator [1 + 5 = 6 Marks]**Problem Statement (2082 Bhadra, Q5)**

What is magnitude comparator? Design a 3-bit magnitude comparator.

A magnitude comparator compares two binary words $A = A_2A_1A_0$ and $B = B_2B_1B_0$ to determine whether $A > B$, $A = B$, or $A < B$.

Design Equations: Define equality condition per bit $x_i = A_i \odot B_i = A_iB_i + \bar{A}_i\bar{B}_i$:

$$\begin{aligned}(A = B) &= x_2 \cdot x_1 \cdot x_0 \\(A > B) &= A_2\bar{B}_2 + x_2A_1\bar{B}_1 + x_2x_1A_0\bar{B}_0 \\(A < B) &= \bar{A}_2B_2 + x_2\bar{A}_1B_1 + x_2x_1\bar{A}_0B_0\end{aligned}$$

Verified Final Result

$$\begin{aligned}(A = B) &= x_2x_1x_0, \quad (A > B) = A_2\bar{B}_2 + x_2A_1\bar{B}_1 + x_2x_1A_0\bar{B}_0 \\(A < B) &= \bar{A}_2B_2 + x_2\bar{A}_1B_1 + x_2x_1\bar{A}_0B_0\end{aligned}$$

2.0.6 Question 6: Full Adder Using Multiplexers [4 Marks]**Problem Statement (2082 Bhadra, Q6)**

Implement a full adder circuit using Multiplexer.

Full Adder has inputs A, B, C_{in} and outputs $\text{Sum} = \sum m(1, 2, 4, 7)$ and $C_{out} = \sum m(3, 5, 6, 7)$.

Using two 4:1 MUX (Select Lines $S_1 = A, S_0 = B$):

$$\begin{aligned}\text{For Sum MUX: } I_0 &= C_{in}, \quad I_1 = \bar{C}_{in}, \quad I_2 = \bar{C}_{in}, \quad I_3 = C_{in} \\ \text{For } C_{out} \text{ MUX: } I_0 &= 0, \quad I_1 = C_{in}, \quad I_2 = C_{in}, \quad I_3 = 1\end{aligned}$$

Verified Final Result

Sum and C_{out} realized with two 4:1 multiplexers sharing select inputs A and B .

2.0.7 Question 7: Combinational vs. Sequential & SR-to-JK Conversion [2 + 5 = 7 Marks]

Problem Statement (2082 Bhadra, Q7)

Differentiate between combinational and sequential circuits. Convert SR flip-flop into JK flip-flop.

SR to JK Flip-Flop Conversion Table:

J	K	$Q(t)$	$Q(t+1)$	S	R
0	0	0	0	0	X
0	0	1	1	X	0
0	1	0	0	0	X
0	1	1	0	0	1
1	0	0	1	1	0
1	0	1	1	X	0
1	1	0	1	1	0
1	1	1	0	0	1

K-Map for S and R Inputs:

$$S = J\bar{Q}, \quad R = KQ$$

Verified Final Result

$$S = J\bar{Q}, \quad R = KQ$$

2.0.8 Question 8: Shift Registers & Johnson's Counter [2 + 3 = 5 Marks]

Problem Statement (2082 Bhadra, Q8)

Write briefly about types of shift registers. Explain the operation of a Johnson's Counter.

Johnson Counter (Twisted Ring Counter): A 4-bit Johnson counter connects the inverted output ($\bar{Q}_3$) of the last flip-flop back to the input (D_0) of the first flip-flop.

- Produces $2n = 8$ distinct states:

$$0000 \rightarrow 1000 \rightarrow 1100 \rightarrow 1110 \rightarrow 1111 \rightarrow 0111 \rightarrow 0011 \rightarrow 0001 \rightarrow 0000$$

- Requires only n flip-flops to count $2n$ states, easily decoded using 2-input AND gates without glitching.

Verified Final Result

Johnson counter modulus is $2n = 8$ states for 4 flip-flops.

2.0.9 Question 9: 3-Bit Ripple UP Counter [5 Marks]

Problem Statement (2082 Bhadra, Q9)

Describe the operation of 3-bit ripple up counter with positive edge triggered clock.

For positive-edge triggered T/JK flip-flops, an UP counter is achieved by clocking each subsequent flip-flop from the inverted output ($\bar{Q}$) of the preceding stage ($CLK_{i+1} = \bar{Q}_i$).

- Counts: 000 $\rightarrow$ 001 $\rightarrow$ 010 $\rightarrow$ 011 $\rightarrow$ 100 $\rightarrow$ 101 $\rightarrow$ 110 $\rightarrow$ 111 $\rightarrow$ 000.

Verified Final Result

Modulus = $2^3 = 8$. Each flip-flop toggles on the positive clock transition of $\bar{Q}_{prev}$.

2.0.10 Question 10: Synchronous Mod-6 Counter Using SR Flip-Flops [7 Marks]

Problem Statement (2082 Bhadra, Q10)

Design the synchronous mod-6 counter using SR flip-flop and draw its timing diagram.

State & SR Excitation Table (0 $\rightarrow$ 1 $\rightarrow$ 2 $\rightarrow$ 3 $\rightarrow$ 4 $\rightarrow$ 5 $\rightarrow$ 0):

Q_2	Q_1	Q_0	Q_2^+	Q_1^+	Q_0^+	S_2	R_2	S_1	R_1	S_0	R_0
0	0	0	0	0	1	0	X	0	X	1	0
0	0	1	0	1	0	0	X	1	0	0	1
0	1	0	0	1	1	0	X	X	0	1	0
0	1	1	1	0	0	1	0	0	1	0	1
1	0	0	1	0	1	X	0	0	X	1	0
1	0	1	0	0	0	0	1	0	X	0	1

K-Map Simplifications:

$$S_0 = \bar{Q}_0, \quad R_0 = Q_0$$

$$S_1 = \bar{Q}_2 \bar{Q}_1 Q_0, \quad R_1 = Q_1 Q_0$$

$$S_2 = Q_1 Q_0, \quad R_2 = Q_0$$

Verified Final Result

$$S_0 = \bar{Q}_0, \quad R_0 = Q_0, \quad S_1 = \bar{Q}_2 \bar{Q}_1 Q_0, \quad R_1 = Q_1 Q_0, \quad S_2 = Q_1 Q_0, \quad R_2 = Q_0$$

2.0.11 Question 11: Synchronous FSM Sequence Detector '110' [7 Marks]

Problem Statement (2082 Bhadra, Q11)

Design a synchronous sequential machine that has 1-bit serial input X and output Z which will be HIGH when the input contains the message 110 (Use SR Flip-Flop).

State Equations & Excitation: Using states $S_0(00)$, $S_1(01)$, $S_2(10)$ and SR flip-flops A, B :

$$\begin{aligned}S_A &= BX, & R_A &= \bar{X} \\S_B &= \bar{A}\bar{B}X, & R_B &= B\bar{X} + A \\Z &= A\bar{X}\end{aligned}$$

Verified Final Result

$$S_A = BX, \quad R_A = \bar{X}, \quad S_B = \bar{A}\bar{B}X, \quad R_B = B\bar{X} + A, \quad Z = A\bar{X}$$

2.0.12 Question 12: Frequency Counter Short Note [3 Marks]

Problem Statement (2082 Bhadra, Q12)

Write a short note on frequency counter.

A digital frequency counter measures the frequency of an unknown periodic input signal by counting the number of cycles that pass through an AND gate enabled for an exact known time period (e.g. 1 second derived from a crystal oscillator and decade dividers).

- **Basic Formula:** $f = \frac{N}{T_{\text{gate}}}$, where N is total counted pulses, T_{gate} is the gate window duration.

Chapter 3

2082 Baishakh Examination Solutions

3.0.1 Question 1: ASCII Code & 2's Complement Addition [1 + 3 = 4 Marks]

Problem Statement (2082 Baishakh, Q1)

Define an ASCII code. Use 2's complement method to perform $(-28+12)_{10}$ in 8-bit signed number representation.

ASCII Code: American Standard Code for Information Interchange is a standard 7-bit character encoding system representing 128 characters (letters, numbers, punctuation, control codes).

2's Complement Addition of $(-28 + 12)_{10}$:

- $+28_{10} = 0001\ 1100_2 \Rightarrow -28_{10} = 1110\ 0011 + 1 = 1110\ 0100_2$
- $+12_{10} = 0000\ 1100_2$

Perform addition:

$$\begin{array}{r} 1110\ 0100\ (-28) \\ +\ 0000\ 1100\ (+12) \\ \hline 1111\ 0000\ (\text{Result is Negative}) \end{array}$$

Magnitude verification: $-(2\text{'s complement of } 1111\ 0000) = -(0000\ 1111 + 1) = -0001\ 0000_2 = -16_{10}$.

Verified Final Result

$$(-28 + 12)_{10} = 1111\ 0000_2 = -16_{10}$$

3.0.2 Question 2: Boolean Algebra Proofs [2 + 2 = 4 Marks]

Problem Statement (2082 Baishakh, Q2)

Prove the following Boolean identities:

- $AB + \bar{A}C + BC = AB + \bar{A}C$ (Consensus Theorem)
- $AB + C(A \oplus B) = BC + A(B + C)$

Proof of (a):

$$\text{LHS} = AB + \bar{A}C + BC(A + \bar{A}) = AB + \bar{A}C + ABC + \bar{A}BC = AB(1 + C) + \bar{A}C(1 + B) = AB + \bar{A}C = \text{RHS}$$

Proof of (b):

$$\text{LHS} = AB + C(\bar{A}\bar{B} + \bar{A}B) = AB + \bar{A}\bar{B}C + \bar{A}BC$$

$$\text{RHS} = BC + AB + AC = AB + BC(A + \bar{A}) + AC(B + \bar{B}) = AB + ABC + \bar{A}BC + ABC + A\bar{B}C = AB + \bar{A}\bar{B}C + \bar{A}BC = \text{LHS}$$

Verified Final Result

Both identities are algebraically verified.

3.0.3 Question 3: Decoder Definition & Octal Priority Encoder [2 + 6 = 8 Marks]

Problem Statement (2082 Baishakh, Q3)

What is a decoder? Design an octal priority encoder with neat circuit diagram.

Octal Priority Encoder (8-to-3): Gives priority to the highest-indexed active input ($D_7 > D_6 > \dots > D_0$).

$$A = D_4 + D_5 + D_6 + D_7$$

$$B = D_2\bar{D}_4\bar{D}_5 + D_3\bar{D}_4\bar{D}_5 + D_6 + D_7$$

$$C = D_1\bar{D}_2\bar{D}_4\bar{D}_6 + D_3\bar{D}_4\bar{D}_6 + D_5\bar{D}_6 + D_7$$

$$V = D_0 + D_1 + D_2 + D_3 + D_4 + D_5 + D_6 + D_7 \quad (\text{Valid Bit})$$

3.0.4 Question 4: Function Realization Using 3x8 Decoder [3 + 3 = 6 Marks]

Problem Statement (2082 Baishakh, Q4)

Realize $X(A, B, C, D) = \sum m(0, 2, 3, 7, 8, 10, 11, 14, 15)$ using a single 3x8 decoder and simplify implementing K-Map.

K-Map Minimization:

$$X(A, B, C, D) = \bar{B}\bar{D} + \bar{B}C + CD + ABC$$

Realization with 3x8 Decoder: Connect A, B, C to select lines S_2, S_1, S_0 , and express the outputs in terms of D : Connect decoder outputs to an OR gate with appropriate $D/\bar{D}$ gating.

3.0.5 Question 5: Synchronous 2-Bit UP/DOWN Counter Using T Flip-Flops [7 Marks]

Problem Statement (2082 Baishakh, Q5)

Design a synchronous 2-bit up/down counter using T flip-flops.

Design Table ($M = 0 \Rightarrow$ UP Counter, $M = 1 \Rightarrow$ DOWN Counter):

M	Q_1	Q_0	Q_1^+	Q_0^+	T_1	T_0
0	0	0	0	1	0	1
0	0	1	1	0	1	1
0	1	0	1	1	0	1
0	1	1	0	0	1	1
1	0	0	1	1	1	1
1	0	1	0	0	0	1
1	1	0	0	1	1	1
1	1	1	1	0	0	1

K-Map Simplification:

$$T_0 = 1$$

$$T_1 = \bar{M}Q_0 + M\bar{Q}_0 = M \oplus Q_0$$

Verified Final Result

$$\mathbf{T_0 = 1, \quad T_1 = M \oplus Q_0}$$

3.0.6 Question 6: 4-Bit PISO Shift Register [3 + 2 = 5 Marks]

Problem Statement (2082 Baishakh, Q6)

Explain the operation of 4-bit parallel-in serial-out (PISO) shift register with necessary circuit and timing diagram for 1101 input data.

A Parallel-In Serial-Out (PISO) shift register loads all 4 bits in parallel during a single clock cycle when $\text{Shift}/\overline{\text{Load}} = 0$, and then shifts the bits out serially bit-by-bit when $\text{Shift}/\overline{\text{Load}} = 1$.

- Data inputs: $D_3D_2D_1D_0 = 1101$.
- After Load: $Q_3 = 1, Q_2 = 1, Q_1 = 0, Q_0 = 1$.
- Serial output bit stream: $1 \rightarrow 0 \rightarrow 1 \rightarrow 1$.

3.0.7 Question 7: Mod-12 Asynchronous Counter with Positive Edge Triggering [3 + 3 = 6 Marks]

Problem Statement (2082 Baishakh, Q7)

Sketch the circuit diagram of mod-12 asynchronous counter having positive-edge triggering clock system implementing JK flip-flops with neat timing diagram.

Mod-12 counter counts 0 through 11 (0000 to 1011), and resets upon reaching 12 (1100₂, where $Q_3 = 1, Q_2 = 1, Q_1 = 0, Q_0 = 0$).

- For positive-edge triggered UP counter, clock each stage from $\bar{Q}$ of previous stage: $\text{CLK}_{i+1} = \bar{Q}_i$.
- Reset condition: Connect Q_3 and Q_2 to a 2-input NAND gate feeding the active-low $\overline{\text{CLR}}$ inputs of all 4 flip-flops:

$$\overline{\text{CLR}} = \overline{Q_3 \cdot Q_2}$$

Verified Final Result

Reset Logic: $\overline{\text{CLR}} = \overline{Q_3 \cdot Q_2}$. Flip-flops count from 0000 to 1011 (12 states).

3.0.8 Question 8: Sequential Machine Sequence Detector '010' Using D Flip-Flops [10 Marks]

Problem Statement (2082 Baishakh, Q8)

Design a sequential machine that consists of one input X and one output Z . $Z = 1$ whenever it detects the serial sequence 010. Implement only D flip-flops.

State Assignment (Mealy Model):

- $S_0(00)$: Reset state.
- $S_1(01)$: Sequence '0' detected.
- $S_2(10)$: Sequence '01' detected.

State Table & D Flip-Flop Inputs ($D = Q^+$):

A	B	X	A^+	B^+	Z	D_A	D_B
0	0	0	0	1	0	0	1
0	0	1	0	0	0	0	0
0	1	0	0	1	0	0	1
0	1	1	1	0	0	1	0
1	0	0	0	1	1	0	1
1	0	1	0	0	0	0	0

K-Map Simplification:

$$D_A = BX$$

$$D_B = \bar{X}$$

$$Z = A\bar{X}$$

Verified Final Result

$$D_A = BX, \quad D_B = \bar{X}, \quad Z = A\bar{X}$$

3.0.9 Question 9: CMOS NAND Gate & TTL Characteristics [4 + 2 = 6 Marks]

Problem Statement (2082 Baishakh, Q9)

Draw the circuit diagram of two-input CMOS NAND gate, explain its logic operation, and list the characteristics of TTL logic family.

CMOS NAND Gate:

- **Pull-Up Network (PUN):** Two PMOS transistors (P_1, P_2) connected in **parallel** between V_{DD} and Output Y .
- **Pull-Down Network (PDN):** Two NMOS transistors (N_1, N_2) connected in **series** between Output Y and Ground.
- **Operation:**
 - If $A = 0$ or $B = 0$, at least one PMOS conducts, pulling output to V_{DD} ('1').
 - If $A = 1$ and $B = 1$, both NMOS conduct in series, pulling output to GND ('0').

TTL Characteristics: $V_{CC} = 5V$, Propagation delay $\approx 10ns$, Power dissipation $\approx 10mW/gate$, Fan-out ≈ 10 .

3.0.10 Question 10: Time Interval Measurement Circuit [4 Marks]**Problem Statement (2082 Baishakh, Q10)**

With the help of functional diagram explain the operation of time measuring circuit.

A digital time interval meter measures the elapsed time Δt between a START pulse and a STOP pulse.

- The START pulse sets an RS flip-flop, enabling an AND gate.
- Precise standard clock pulses from a crystal oscillator (e.g. 1MHz, period $T = 1\mu s$) pass through the open gate into a digital counter.
- The STOP pulse resets the flip-flop, disabling the gate.
- $\Delta t = N \times T_{clk}$, where N is the total registered counts.

Chapter 4

2081 Ashwin Examination Solutions

4.0.1 Question 1: Merits & Demerits of Digital Signals [2 Marks]

Problem Statement (2081 Ashwin, Q1)

Mention merits and demerits of digital signal over analog signal.

Merits: High noise immunity, easy error detection/correction, perfect reproduction, simple digital IC integration and programmability.

Demerits: Higher bandwidth requirement, quantization error/noise during analog-to-digital conversion, and requirement for high-speed converters.

4.0.2 Question 2: Radix Gray Code Conversions [3 Marks]

Problem Statement (2081 Ashwin, Q2)

Perform: (a) $(6E.2C)_{16} = (?)_8$, (b) $(10110110)_{\text{Gray}} = (?)_2$.

Part (a): $(6E.2C)_{16} \rightarrow \text{Octal}$

- Hex to Binary: $6 = 0110, E = 1110, .2 = 0010, C = 1100 \implies 01101110.00101100_2$
- Regroup into 3-bit octal groups: $001\ 101\ 110\ .\ 001\ 011\ 000_2 = (156.130)_8 = (\mathbf{156.13})_8$.

Part (b): $(10110110)_{\text{Gray}} \rightarrow \text{Binary}$

$$b_7 = 1, b_6 = 1 \oplus 0 = 1, b_5 = 1 \oplus 1 = 0, b_4 = 0 \oplus 1 = 1, b_3 = 1 \oplus 0 = 1, b_2 = 1 \oplus 1 = 0, b_1 = 0 \oplus 1 = 1, b_0 = 1 \oplus 0 = 1$$

Result: $(\mathbf{11011011})_2$.

Verified Final Result

$$(a) (6E.2C)_{16} = (\mathbf{156.13})_8 \quad (b) (10110110)_{\text{Gray}} = (\mathbf{11011011})_2$$

4.0.3 Question 3: BCD to 7-Segment 'f' Segment Decoder [4 Marks]

Problem Statement (2081 Ashwin, Q3)

Design the simplest logic circuit for 'f' segment for the BCD-to-seven segment display decoder.

The 'f' segment is active (HIGH) for decimal digits: 0, 4, 5, 6, 8, 9. Inputs 10 to 15 are Don't Cares (d).

$AB \backslash CD$	00	01	11	10
00	1 (m_0)	0 (m_1)	0 (m_3)	0 (m_2)
01	1 (m_4)	1 (m_5)	0 (m_7)	1 (m_6)
11	X	X	X	X
10	1 (m_8)	1 (m_9)	X	X

K-Map Simplification:

Group 1 (Row 10, 11): A
 Group 2 (m_0, m_4, d_{12}, m_8): $\bar{C}\bar{D}$
 Group 3 (m_4, m_5, d_{12}, d_{13}): $B\bar{C}$
 Group 4 (m_5, m_6, d_{13}, d_{14}): $B\bar{D}$

Verified Final Result

$$f = A + \bar{C}\bar{D} + B\bar{C} + B\bar{D}$$

4.0.4 Question 4: 4:1 MUX Realization & Combined Half Adder/Subtractor [3 + 7 = 10 Marks]

Problem Statement (2081 Ashwin, Q4)

Implement $Y(A, B, C) = \sum m(0, 2, 3, 5, 7)$ using only a single 4:1 MUX. Design a combined Half-Adder and Half-Subtractor with a mode control.

Part 1: 4:1 MUX Realization (Select Lines $S_1 = A, S_0 = B$):

- $AB = 00 \Rightarrow m_0, m_1 \Rightarrow Y = \bar{C} \text{ (} I_0 = \bar{C} \text{)}$
- $AB = 01 \Rightarrow m_2, m_3 \Rightarrow Y = 1 \text{ (} I_1 = 1 \text{)}$
- $AB = 10 \Rightarrow m_4, m_5 \Rightarrow Y = C \text{ (} I_2 = C \text{)}$
- $AB = 11 \Rightarrow m_6, m_7 \Rightarrow Y = C \text{ (} I_3 = C \text{)}$

Part 2: Combined Half-Adder / Half-Subtractor (Mode $M = 0$ for Add, $M = 1$ for Sub):

- **Sum / Difference:** $S/D = A \oplus B$ (Identical for both).
- **Carry / Borrow:** $C_{\text{out}} = AB$ (when $M = 0$), $B_{\text{out}} = \bar{A}B$ (when $M = 1$).
- **Unified Output:** $C/B = (A \oplus M) \cdot B$.

Verified Final Result

MUX Inputs: $I_0 = \bar{C}, I_1 = 1, I_2 = C, I_3 = C$.

Combined Adder/Subtractor Logic: Sum/Diff = $A \oplus B$, Carry/Borrow = $(A \oplus M) \cdot B$.

4.0.5 Question 5: Mod-6 Synchronous DOWN Counter Using JK Flip-Flops [7 Marks]

Problem Statement (2081 Ashwin, Q5)

Design a mod-6 synchronous down counter using JK flip-flops ($5 \rightarrow 4 \rightarrow 3 \rightarrow 2 \rightarrow 1 \rightarrow 0 \rightarrow 5$).

State & JK Excitation Table:

Q_2	Q_1	Q_0	Q_2^+	Q_1^+	Q_0^+	J_2	K_2	J_1	K_1	J_0	K_0
1	0	1	1	0	0	X	0	0	X	X	1
1	0	0	0	1	1	X	1	1	X	1	X
0	1	1	0	1	0	0	X	X	0	X	1
0	1	0	0	0	1	0	X	X	1	1	X
0	0	1	0	0	0	0	X	0	X	X	1
0	0	0	1	0	1	1	X	0	X	1	X

K-Map Simplifications:

$$J_0 = 1, \quad K_0 = 1$$

$$J_1 = \bar{Q}_2 \bar{Q}_0, \quad K_1 = \bar{Q}_0$$

$$J_2 = \bar{Q}_1 \bar{Q}_0, \quad K_2 = \bar{Q}_1 \bar{Q}_0$$

Verified Final Result

$$J_0 = 1, \quad K_0 = 1, \quad J_1 = \bar{Q}_2 \bar{Q}_0, \quad K_1 = \bar{Q}_0, \quad J_2 = K_2 = \bar{Q}_1 \bar{Q}_0$$

4.0.6 Question 6: 4-Bit SIPO Shift Register with Data 1011 [3 + 3 = 6 Marks]

Problem Statement (2081 Ashwin, Q6)

Explain the operation of 4-bit serial-in parallel-out (SIPO) shift register with necessary circuit and timing diagram for input data of 1011.

Serial input sequence 1011 (MSB first: 1, 0, 1, 1):

Clock	Serial In	Q_3	Q_2	Q_1	Q_0
Initial	—	0	0	0	0
1st CLK	1	1	0	0	0
2nd CLK	0	0	1	0	0
3rd CLK	1	1	0	1	0
4th CLK	1	1	1	0	1

4.0.7 Question 7: 3-Bit Asynchronous UP/DOWN Counter [4 + 3 = 7 Marks]

Problem Statement (2081 Ashwin, Q7)

Describe briefly the operation of 3 bit up/down asynchronous counter having negative edge triggering clock system with neat circuit diagram and timing diagram.

For negative-edge triggered T/JK flip-flops:

- **UP Counting:** Clock next stage from normal output Q ($\text{CLK}_{i+1} = Q_i$).
- **DOWN Counting:** Clock next stage from inverted output $\bar{Q}$ ($\text{CLK}_{i+1} = \bar{Q}_i$).
- **Mode Control (M):** Using steering logic: $\text{CLK}_{i+1} = \bar{M}Q_i + M\bar{Q}_i$.

4.0.8 Question 8: Sequence Detector '101' Using SR Flip-Flops [10 Marks]

Problem Statement (2081 Ashwin, Q8)

Design a sequential machine that consists of one input X and one output Z . $Z = 1$ whenever it detects the serial sequence 101. Implement only SR flip-flops.

State Table & SR Inputs ($S_0 = 00, S_1 = 01, S_2 = 10$):

$$\begin{aligned} S_A &= BX, & R_A &= \bar{X} \\ S_B &= \bar{A}\bar{B}X + AX, & R_B &= B\bar{X} \\ Z &= AX \end{aligned}$$

Verified Final Result

$$S_A = BX, \quad R_A = \bar{X}, \quad S_B = X(\bar{A}\bar{B} + A), \quad R_B = B\bar{X}, \quad Z = AX$$

4.0.9 Question 9: TTL NOR Gate CMOS Characteristics [5 + 2 = 7 Marks]

Problem Statement (2081 Ashwin, Q9)

Draw the circuit diagram of two-input TTL NOR gate, explain its operation, and list the characteristics of CMOS logic family.

TTL NOR Gate Operation: Uses multi-emitter input transistors and totem-pole output stage (Q_3 pull-up transistor, Q_4 pull-down transistor). When either input is HIGH, the corresponding phase-splitter and pull-down transistor conduct, driving output LOW.

CMOS Logic Family Characteristics: Extremely low static power dissipation ($< 10\text{nW}$), high noise margin ($V_{DD}/2$), wide operating voltage range ($3\text{V} - 15\text{V}$), high fan-out (> 50).

4.0.10 Question 10: Frequency Measurement Using Digital Counter [4 Marks]**Problem Statement (2081 Ashwin, Q10)**

With the help of functional diagram explain the operation of frequency measurement.

Functional Operation:

1. **Input Amplifier / Schmitt Trigger:** Converts arbitrary periodic analog input into sharp rectangular digital pulses.
2. **Time Base Generator:** Crystal oscillator + decade frequency dividers produce an accurate gating window pulse T_g (e.g. 1.000s).
3. **Main Gate (AND Gate):** Passes input signal pulses to the decade counter only while T_g is active.
4. **Display Unit:** Displays the total count N , which directly equals frequency $f = \frac{N}{T_g}$ Hz.

Verified Final Result

$$f_x = \frac{N}{T_g}$$